

Controlled Evaluation of Class-Imbalance Mitigation Across Histopathology Patch Classification and WSI-Level MIL

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Abstract

This study evaluates methods for mitigating class imbalance in computational pathology, covering patch classification and whole-slide multiple-instance learning (MIL) in CAMELYON16, TCGA-UT, BRACS, and PANDA, all represented with frozen Virchow2 features. Patient- or case-disjoint splits are created before imbalance is manipulated, and only training support changes: a full natural anchor is accompanied by size-matched balanced, moderate ($\rho = 10$), and severe ($\rho = 100$ where feasible) conditions. Validation and test split units, held-out prevalence, training-set size, model family, and numbers of optimizer updates remain fixed. Patch conditions retain a common patient/case and slide pool while changing patch allocation; MIL conditions vary slide support and report the resulting change in independent-unit diversity. The primary discrimination endpoint is balanced accuracy, evaluated alongside a co-primary tail-calibration endpoint. Mitigation recovery is assessed only when prespecified thresholds show a meaningful imbalance-induced deficit on the corresponding endpoint. Secondary analyses address classwise discrimination, tail-sensitive calibration, computational cost, intrinsic separability, and condition-specific learnability.

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1 Introduction

Class imbalance changes the empirical objective faced by a classifier: frequent classes supply more optimization terms, while rare classes may contribute too little evidence to establish stable decision boundaries. In pathology, the problem occurs at two distinct prediction units. Patch classification uses directly labelled crops, whereas WSI-level MIL predicts from bags whose instances inherit only a weak slide label. The same slide cohort can therefore be nearly balanced between diagnoses while containing very sparse class-informative tissue. Aggregate accuracy does not isolate either failure mode, motivating class-balanced discrimination and probability-quality evaluation (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Native-dataset comparisons do not identify an imbalance effect by themselves. They simultaneously vary task, class separability, training support, label granularity, and evaluation prevalence. This study instead changes class allocation only in the training partition and retains identical natural validation and test cohorts. A size-matched balanced CE condition estimates attainable performance at the same training budget, so mitigation can be interpreted as recovery of damage attributable to the imposed imbalance rather than as an unconstrained gain over a convenient baseline.

The study addresses three research questions:

RQ1. Effect of imbalance. At fixed task, split-unit partition, encoder, training-set size, and evaluation set, how does increasing training imbalance affect tail-class discrimination and calibration in patch classification and WSI-level MIL?

RQ2. Mitigation effectiveness. Which mitigation methods recover performance lost specifically to training imbalance, and what are their calibration and computational costs?

RQ3. Predictors of damage and mitigation headroom. Can small exploratory models help explain why some controlled comparisons are more vulnerable than others? Specifically, are achieved imbalance severity and support–difficulty alignment associated with whether damage occurs, how large it is, and how much is recovered after damage? Separability, learnability, and support measures are examined as sensitivity predictors.

RQ1 establishes whether a recoverable problem exists, and RQ2 asks which methods address that demonstrated deficit. RQ3 is an exploratory association analysis rather than another method ranking: it relates prespecified properties of each comparison to damage occurrence, damage magnitude, and, where damage is demonstrated, recovery. The datasets and preprocessing provenance are documented in the companion dataset report, and all detailed method objectives and variants are defined in the companion methods report (Queisler, 2026a,b).

2 Experimental Design

A controlled imbalance experiment is informative only when the balanced and imbalanced models differ in their training class allocation rather than in their data source, representation, or evaluation cohort. The design therefore establishes the split-unit partition, frozen features, and held-out examples once, then constructs several training distributions at a common support budget. The following subsections define the two prediction regimes, explain how their evidence is controlled, and specify how mitigation methods are trained and compared.

Readers seeking a compact visual entry point can consult the benchmark overview in Appendix A (Figure 6); the subsections below provide the formal specification.

2.1 Datasets and prediction regimes

Table 1 defines the prediction example and label in each available dataset–regime. The terms *patch* and *tile* both denote a 256×256 image crop; *patch* is used throughout this report. A patch-level example is one such crop with the label specified in the second column. A bag-level example is one WSI represented by frozen embeddings of its eligible patches and receives one label for the entire WSI.

Every eligible patch embedding enters the MIL model for its WSI, so bag size reflects the tissue that remains after the dataset-specific eligibility pipeline. The complete bag for a given slide is identical across training conditions and methods.

Before examples are selected, each *split unit*—a patient or participant where available and a biopsy/WSI otherwise—is assigned to exactly one partition: the training, validation, or test cohort. All slides from the same split unit follow that assignment. Dataset–regime eligibility is also fixed before splitting. A sample is excluded if it lacks a valid target or split-unit identifier, yields no patch under the dataset-specific tissue rule, lacks an annotation required to assign a patch label, or lacks readable frozen features. The same rules are applied before constructing every imbalance condition.

Dataset	Patch-level example and label	MIL bag and bag label	Split unit
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	One WSI bag; metastasis-bearing (tumor) versus normal slide	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI’s cancer type	One diagnostic-WSI bag; the same one of 32 cancer types	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	One WSI bag; official seven-class WSI label defined by the most aggressive lesion detected in the slide	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	One biopsy-WSI bag; ordinal ISUP grade 0–5	Biopsy (one WSI)

Table 1: Dataset-specific definitions of patch-level examples and bag-level examples. Only TCGA-UT uses the same target in both regimes. CAMELYON16 and PANDA change diagnostic granularity, while BRACS changes from the annotated ROI subtype to the official most-aggressive-lesion WSI subtype.

Patch classification. Eligible patches are embedded once with frozen Virchow2. Concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). For CAMELYON16 and PANDA, a patch label is derived from the spatially aligned expert mask and can differ from the WSI label. A TCGA-UT patch inherits the cancer-type label of the diagnostic WSI from which the dataset authors cropped it. A BRACS patch inherits the subtype of its pathologist-annotated ROI. All patches that pass these dataset-specific eligibility and label rules enter the eligible pool before condition-specific training subsampling. The common classifier is a two-layer multilayer perceptron (MLP) with a 512-unit hidden representation, rectified linear unit (ReLU) activation, dropout, and a linear class output. CFAL alone replaces the linear output with a prototype-based affinity head; OKO adds a training-only auxiliary head that is discarded for evaluation.

WSI-level MIL. A slide is one bag of frozen patch embeddings; here, an *instance* means one patch representation inside that bag, not a separately labelled training example. CAMELYON16 bags use the slide diagnosis, TCGA-UT bags use the diagnostic WSI’s cancer type,

and PANDA bags use the biopsy’s ordinal ISUP grade. A BRACS bag uses the official seven-class WSI annotation, which the dataset defines from the most aggressive lesion detected in the slide. This label is read from the official WSI metadata rather than reconstructed from ROI annotations, so WSIs without released ROIs remain eligible and no project-specific severity ordering can alter the target. BRACS WSIs are sampled at 20x into non-overlapping 256×256 patches and use the CAMELYON16 foreground stack: at least 10% Otsu foreground, grayscale standard deviation of at least 8, Canny edge evidence, and at least two tissue neighbours. The common attention-MIL model maps every eligible instance to a 256-dimensional space, assigns normalized attention weights, and classifies their weighted average. SC-MIL adds a training projection head for its contrastive objective, and MDE-inspired MIL replaces the single classifier with two expert heads. Slide labels are the only supervision; patch labels, ROI labels, and masks are not used by the primary MIL training runs (Brancati et al., 2022; Queisler, 2026a).

Comparative scope. Only TCGA-UT permits a direct patch-versus-WSI comparison. Both regimes predict the same cancer-type label within the same participant-disjoint cohort. In CAMELYON16, patch classification detects tumor tissue, whereas a WSI bag is classified as metastasis-bearing or normal. In PANDA, patch classification detects cancer tissue, whereas a WSI bag predicts the biopsy’s ISUP grade. In BRACS, a patch predicts its ROI-specific subtype, whereas a WSI bag predicts the official most-aggressive-lesion subtype for the complete slide. The CAMELYON16, PANDA, and BRACS regimes are therefore separate dataset–target groups rather than paired measurements of one task.

2.2 Controlled comparison

The construction asks a simple counterfactual question: how would the same prediction task perform if a fixed training budget were distributed evenly across classes rather than concentrated in a head class? Answering this question requires more than specifying an imbalance ratio. The comparison must first fix the split-unit partition, eligible examples, representation, evaluation cohort, and total controlled training support; only then can class allocation be changed. Figure 1 summarizes this separation between fixed design choices and the intended intervention.

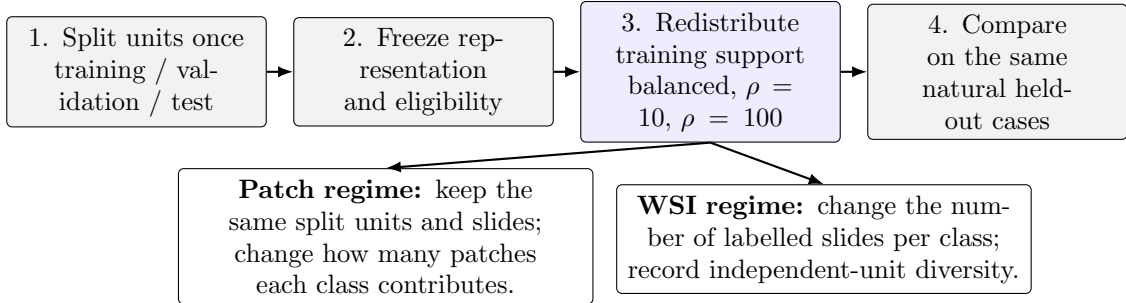


Figure 1: Controlled construction of the benchmark. Gray steps are fixed within a split; the blue step is the intended intervention. Patch imbalance redistributes patches within a common split-unit and slide pool, whereas WSI imbalance changes labelled slide support.

Each eligible dataset–regime–split has four training conditions:

1. the full natural training set, used only as a descriptive anchor;
2. a size-matched balanced training set;
3. a moderate imbalance condition targeting $\rho = 10$; and
4. a severe imbalance condition targeting $\rho = 100$ where feasible.

The balanced, moderate, and severe conditions contain the same total training support T : patches for patch classification and slides for MIL. The shared T is selected jointly with severity, not from size alone: for each requested ratio, construction first finds the largest ratio the training data can realize at any candidate T , then fixes T to the largest value attaining those maxima simultaneously for every locked tail assignment, subject to the support constraints defined below. A larger T chosen without regard to severity can flatten the realized profile toward the balanced condition even when a smaller T would have supported the intended contrast, which would defeat the comparison the three conditions exist to make. The natural anchor may contain a different total and is excluded from the imbalance deficit and recovery estimands in Section 3.4. Thus contrasts among the three controlled conditions change class allocation without also changing the number of training examples.

The target ratio $\rho_{\text{target}} \in \{1, 10, 100\}$ specifies the intended intervention, but integer allocation and finite inventories can change the realized class distribution. The achieved ratio is therefore recomputed from every manifest. For realized class supports $N_1, \dots, N_K$, it is

$$\rho_{\text{achieved}} = \frac{N_{\max}}{N_{\min}}. \quad (1)$$

This quantity records the extreme head–tail contrast. The complementary normalized-entropy deficit

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^K p_c \log p_c}{\log K}, \quad p_c = \frac{N_c}{\sum_j N_j}. \quad (2)$$

uses all classes and is normalized for class count: it is zero for a balanced allocation and increases as support becomes concentrated. Both quantities are computed separately at patch and slide level from each realized training manifest. Per-class split-unit, slide, and patch counts are reported alongside them, because neither scalar reveals whether many patches originate from only a few cases.

Split-unit-disjoint training, natural-validation, and natural-test partitions are created first. Imbalance manipulation is confined to training support. Within a split, every condition uses exactly the same validation and test cases and eligible held-out examples. No held-out example is deleted, duplicated, or reweighted according to the training severity. Patch-level evaluation uses every eligible held-out patch, and slide-level evaluation uses every eligible held-out slide. Consequently, recall is estimated from all available held-out cases and prevalence-dependent precision and macro F1 are compared at one fixed test prevalence.

The validation cohort also retains its natural distribution. It is used for hyperparameter and checkpoint selection but never resized to resemble a training condition. Training, validation, and test manifests are versioned with split-unit identifiers, class counts, construction seed, and content hashes before model fitting.

2.3 Independent evidence and support floors

A patch is a training example, but it is not independent evidence when thousands of patches come from the same case or slide. In this report, *independent support* therefore means the number of distinct split units and slides that underlie a class’s training examples. Controlling this support prevents an apparent patch-level head–tail contrast from being merely a contrast between many correlated crops from one case and fewer crops from another.

The rule is adapted to the prediction regime. For patch classification, a fixed per-class split-unit pool and, within it, a fixed slide pool are made available to the balanced and all imbalanced conditions: every condition draws its patches only from this pool, and the condition with the largest patch allocation for a class covers the pool in full, so no unit in it goes unused by every condition together. A smaller condition for the same class, in particular a severe tail count well below that maximum, may leave some pool units untouched; this is the intended shape of an

imbalanced allocation, not a violation of the shared pool. Sampling visits split units, then slides within units, and only then patches. No split unit may supply more than 10% of a class’s patch allocation and no slide more than 5%; deterministic round-robin selection enforces these caps before any unit is revisited. Patch imbalance can therefore change the number of patches per class without changing which split-unit and slide pools are available to draw from. For MIL, a slide is itself the labelled example, so reducing slide support necessarily reduces independent support. Selection visits split units before taking additional slides from the same unit; each slide contributes once, and no unit supplies more than 10% of a class’s slides. MIL contrasts are consequently interpreted as controlled slide-support imbalance, including its induced change in independent-unit diversity. Every manifest reports per-class split-unit, slide, and patch counts, maximum contributions, and the fraction of the eligible independent pool retained.

The smallest defensible tail support is estimated before the definitive comparison with a *support pilot*. These balanced CE runs ask when additional independent training units cease to materially improve natural-validation performance; they do not tune a hyperparameter or contribute observations to the definitive analysis. In the patch regime, one pilot unit is a split unit, or a slide where split unit and slide coincide. Each selected unit contributes the same fixed quota q_{patch} of patches, distributed as evenly as possible over its eligible slides. The quota is the largest value the patient and slide contribution caps admit at every candidate level and every construction ordering, held fixed so increasing a pilot level adds independent units rather than patches per unit. A caps-admissible quota is necessary but not sufficient: a unit cannot supply more patches than it holds, so a unit is eligible for a candidate level only if its own inventory meets the candidate quota, and the quota at a given level is set from that level’s order statistic of per-unit inventory over the eligible units rather than from the scarcest unit in one seeded ordering. A candidate level that no quota can satisfy is dropped from the curve rather than reusing a looser quota from a larger level. In the MIL regime, one pilot unit is a labelled slide represented by all its eligible instances.

Nested pilots compare 10, 15, 20, 30, and 50 units per class, or all available units when the scarcest class has fewer. They are repeated for three independently seeded *construction orderings*. An ordering is a fixed sequence of eligible units; its prefix of length m is simply the first m units in that sequence. Thus the 15-unit pilot retains the same first 10 units as the 10-unit pilot and adds the next five. At lower levels of the hierarchy, every split unit has a fixed slide order and every slide has a fixed patch order. Figure 2 illustrates this nesting.

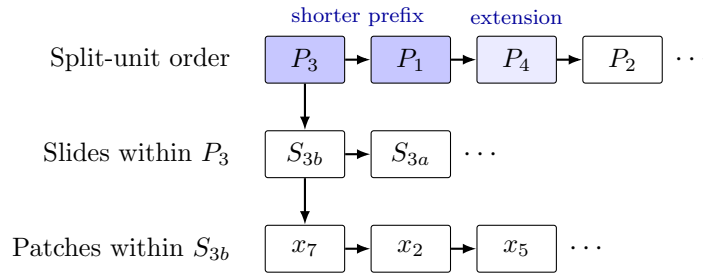


Figure 2: Hierarchy behind one construction ordering. A seed fixes the split-unit sequence and the local slide and patch sequences. The darker unit boxes form a shorter prefix; the lighter third box is added, without replacing earlier units, when the prefix is extended.

Because every larger pilot level extends the preceding prefix, adjacent levels are paired within an ordering and evaluated on the same natural-validation cohort. The smallest level whose next increment changes mean balanced accuracy by less than 0.01 and every class recall by less than 0.02 in all three orderings is the *stability floor*; if none qualifies, the largest candidate is used.¹

¹No pilot level is exempt from the caps. A level is dropped from the curve when no quota satisfies both caps for every construction seed—e.g., the 5% slide cap needs at least 20 contributing slides, so a class whose split

The definitive minimum is the larger of this empirical stability floor and a prespecified design floor. Each class must contain at least 10 split units and 20 slides, or 20 slides when split unit and slide coincide, and any class-aware batch or set must contain at least two distinct same-class units. The 10-unit and 20-slide values are not estimated performance thresholds: they are the smallest counts compatible with the 10% split-unit and 5% slide contribution caps, respectively. The same 20-slide lower bound is retained for MIL to keep slide-based method comparisons outside a five-example regime even though MIL has no separate 5% slide cap. A requested severity is reduced to the largest allocation satisfying both floors; a dataset–regime is excluded if even its balanced condition fails them. Pilot quotas, curves, seeds, and resulting floors are frozen and reported with the definitive manifests.

Before assigning support ranks, the largest balanced pilot level is used to measure class difficulty. For each of the three pilot constructions, a class-balanced linear probe makes five-fold out-of-fold predictions with case/patient-disjoint **StratifiedGroupKFold** partitions. Class difficulty is $d_c = 1 - \overline{\text{recall}}_c$, where recall is averaged across the three constructions; five-neighbour kNN recall is retained as sensitivity evidence. Pilot hashes, folds, classwise scores, and the easiest-to-hardest ranking are signed. Equal scores retain native class order.

2.4 Severity profiles and tail assignments

A binary task needs only a head and a tail count, but a multiclass task also needs a rule for the classes between those extremes. The exponential construction places the K class supports at equal intervals on a logarithmic scale. For a locked order from head ($i = 0$) to tail ($i = K - 1$), it assigns weights

$$w_i(\rho_{\text{target}}) = \rho_{\text{target}}^{-i/(K-1)}, \quad n_i^* = T \frac{w_i(\rho_{\text{target}})}{\sum_{j=0}^{K-1} w_j(\rho_{\text{target}})}, \quad i = 0, \dots, K - 1. \quad (3)$$

so the head-to-tail ratio is exactly ρ_{target} before integer rounding. For example, with four classes and $\rho_{\text{target}} = 100$, the unnormalized weights are approximately 1, 0.215, 0.046, and 0.01; the two body classes lie geometrically between the head and tail instead of being assigned ad hoc counts. Binary targets use $n_{\text{head}}^*/n_{\text{tail}}^* = \rho_{\text{target}}$ directly. The same exponential rule is applied to every multiclass dataset–regime, so a given ρ_{target} denotes the same allocation shape in each of them.

The values n_i^* are real-valued targets, whereas a manifest must contain whole patches or slides. *Constrained integer allocation* converts them into usable class counts. Each target is rounded to the nearest integer and clipped to the class’s support floor and available unique support. If the rounded counts do not sum to T , units are added to or removed from the classes with the largest rounding residuals until the total is exact. The resulting integers n_i therefore satisfy $\sum_i n_i = T$ and every class-specific lower and upper bound while remaining close to the exponential targets under this deterministic adjustment.

Counts determine how many units a class contributes; construction orderings determine which units they are. One master ordering is created per class at each hierarchy level. Every condition selects a deterministic prefix, or a split-unit-round-robin extension of a prefix, from these same orderings. For patch classification, all fixed-pool split units and slides contribute before additional patches are allocated; for MIL, slides are selected by cycling through the split-unit prefix before revisiting a unit.

If a target ratio is infeasible, the same rule is applied to every dataset and regime: use the largest attainable ratio and report the requested ratio, achieved ratio, limiting class, realized counts, and binding support floor. Validation or test support is never reduced to manufacture feasibility. This general fallback is especially relevant to high- K tasks, where an exponential

units mostly contribute a single slide can leave levels below 20 infeasible. Dropped levels are recorded in the frozen pilot report.

profile must allocate support to many intermediate classes as well as the tail; it is not a dataset-specific exception. A stable lower-severity condition is preferred to a nominal $\rho = 100$ condition supported by only a few independent units.

The fallback has a limit: a moderate or severe condition whose realized allocation is indistinguishable from the balanced condition—achieved ratio within a small tolerance of 1, or realized counts identical to the balanced manifest—is not a lower-severity instance of the intended contrast but a null one, since the imbalance intervention did not occur. Such a condition is reported as degenerate rather than fitted and averaged with genuine moderate or severe conditions from other splits or assignments, and it is excluded from the imbalance deficit and recovery estimands in Section 3.4. Its requested ratio, achieved ratio, limiting class, and binding support floor are recorded alongside the exclusion, using the same fields every condition already reports.

The profile determines support by rank, so a second decision maps semantic classes to those ranks. Using only one mapping would confound scarcity with intrinsic class difficulty. Multiclass targets therefore use native, difficulty-aligned, and difficulty-reversed assignments where distinct: native retains clinical, ordinal, or natural-support order; aligned places easiest classes at the head and hardest classes at the tail, compounding scarcity with difficulty; reversed places hardest classes at the head and easiest classes at the tail, providing compensation. Binary tasks retain native and reversed orientations. A difficulty mapping that duplicates an earlier mapping in any locked split is omitted across that dataset-regime with its reason recorded. These three sensitivity assignments separate interpretable mechanisms but do not exhaust all possible class-support mappings. Figure 3 illustrates the design.

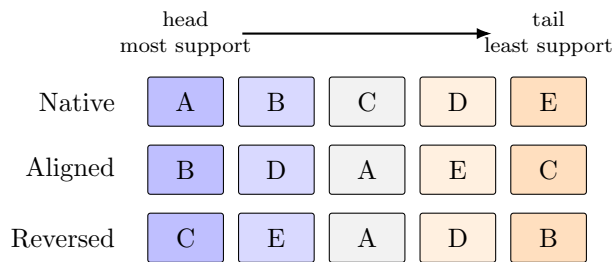


Figure 3: Illustrative tail assignments for a five-class target. Columns are support ranks fixed by the severity profile. Difficulty increases from B to C: aligned compounds difficulty with scarce support, while reversed compensates it.

The random quantities are separated into five seed families so that changing one source of variation does not silently change another. Table 2 defines their roles.

Seed family	Determines	Prespecified use
Split-unit partition	Membership of the training, validation, and test cohorts	Three locked patient- or case-disjoint splits
Tail assignment	Mapping of semantic classes to head–tail support ranks	Native, difficulty-aligned, and difficulty-reversed mappings from signed pilot evidence
Construction	Split-unit, slide, and patch orderings and the selected evidence	Three pilot orderings and a disjoint fourth ordering for definitive manifests
Initialization	Parameter initialization and optimization randomness	Two tuning seeds and five disjoint confirmation seeds
Resampling	Bootstrap draws and permutation swaps	Uncertainty intervals and hypothesis tests only

Table 2: Seed families and their non-overlapping roles in the benchmark.

No seed is chosen based on test performance. The size-matched balanced manifest and its

trained CE baseline are shared by all tail assignments within a split.² The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

2.5 Mitigation methods

Table 3 lists only the methods evaluated in this benchmark. CE and MIL CE are the size-matched no-mitigation references. Method objectives, tested adaptations, ablations, defaults, and fidelity qualifications are defined exclusively in the companion methods report (Queisler, 2026b).

Family	Tested methods	Regime
No-mitigation reference	CE; MIL CE	Patch; WSI
Data-level sampling and augmentation	Balanced sampling; RankMix-inspired feature mixing	Both for sampling; WSI for RankMix-inspired
Losses and prior correction	Weighted CE; focal loss; train-time and post-hoc logit adjustment; CFAL	Both, except CFAL patch only
Hybrid sampling-loss	CE + soft F1; CE + soft MCC; OKO set learning; MDE-inspired dual-expert MIL	Patch for CE hybrids and OKO; WSI for MDE-inspired
Representation and architecture	cRT; SC-MIL	Both for cRT; WSI for SC-MIL

Table 3: Prespecified method roster, using the same intervention-level taxonomy as the companion methods report. “Both” means the patch and WSI implementations defined there, not a comparison of their raw scores.

2.6 Training, selection, and replication

The frozen encoder, classifier family, optimizer family, batch definition, augmentation, and numerical precision are held fixed within a regime. Training is budgeted only in optimizer updates; epochs are neither a stopping rule nor a unit of comparison. Let B be the regime’s locked batch size and T the common controlled support. A single-stage model receives

$$U = 30 \left\lceil \frac{T}{B} \right\rceil \quad (4)$$

updates, corresponding to 30 reference passes through T examples under ordinary sampling. Balanced sampling, replacement, synthetic bags, and dual loaders do not change U . RankMix receives U teacher updates and, after reinitialization, U student updates. cRT receives U stage-one updates and $\lceil 0.2U \rceil$ stage-two classifier updates. MDE performs U joint updates, each consuming its prespecified natural and balanced minibatches. The natural anchor uses the same formula with its own support and is descriptive. Processed examples, unique-example exposures, and updates are all reported. Each run stores validation metrics at fixed update intervals. The selected checkpoint maximizes natural-validation balanced accuracy, with macro F1 and then lower NLL as deterministic tie-breakers. Test predictions are produced once from that locked checkpoint.

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity, and the search adapts within a pre-specified envelope rather than committing to a single fixed grid. Every trainable method draws its learning rate from a common seven-value envelope (Appendix B); the active search window is the four adjacent envelope values used in the fixed-grid design, crossed with the method’s

²At $\rho_{\text{target}} = 1$, every class receives equal support. Permuting semantic classes across support ranks therefore leaves the allocation unchanged, so tail assignment is not a meaningful factor for the balanced condition.

imbalance-specific control where one applies. If the winning learning rate falls on either edge of the active window, that edge value is not accepted as an optimum: the window shifts one position toward the edge, the three overlapping candidates are reused rather than retrained, and only the one new candidate is fitted before the method is reselected. This repeats until the winner is interior to its window or the envelope’s boundary is reached, at which point the boundary value is retained and the axis is recorded as tuning-limited. Focal loss and the two soft-label hybrids, CE + soft F1 and CE + soft MCC, apply the same shifting rule to their auxiliary-strength axis over a dedicated seven-value envelope (Appendix B); every other method’s imbalance-specific control keeps the fixed four-value grid in Table 6, since its domain is naturally bounded. For weighted CE, balanced sampling, focal loss, CE + soft F1, and CE + soft MCC, an imbalance-specific strength of zero is mathematically identical to plain CE at the same learning rate: it applies no class reweighting or sampling adjustment, reduces the focal term to the plain cross-entropy loss, or drops the auxiliary term entirely. That configuration’s metrics are therefore copied from CE’s already-fitted same-learning-rate run rather than retrained, giving each of these five methods one additional comparison point at no extra training cost. Each axis’s search is bounded to at most three rounds by the size of its envelope, and a resolved or tuning-limited axis is never retrained in a later round. CE and methods without a meaningful imbalance-specific control search the adaptive learning-rate envelope alone. Train-time logit adjustment is tuned jointly over τ and learning rate, with τ held to its fixed grid; post-hoc logit adjustment inherits the selected CE model and tunes only τ because it performs no gradient optimization.

Every hyperparameter candidate is fitted with the same two tuning initialization seeds for each prespecified split-unit partition and tail assignment. Selection is performed separately for each method, dataset, prediction regime, and imbalance severity. Within each such selection unit, natural-validation balanced accuracy is averaged across both tuning seeds, all tuning splits, and all tail assignments. The configuration with the highest aggregate value is selected; macro F1 and then lower NLL are deterministic tie-breakers. The chosen configuration is locked before any of the five disjoint confirmation initialization seeds are fitted and before test evaluation. Tuning runs never enter confirmation effect estimates. For cRT, stage one inherits the selected CE configuration, while the stage-two classifier learning rate is selected separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

A *controlled contrast* is the difference between two matched runs, such as balanced CE versus imbalanced CE or a mitigation method versus imbalanced CE. The two runs use the same split-unit partition, tail assignment, confirmation-seed index, and held-out cases, so the contrast does not mix the intended change with a different cohort or replication. Each locked partition and tail assignment is repeated with the five confirmation seeds. Exactly three split-unit partition seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis. The split counts, support floors, condition manifests, tuning and confirmation seeds, method grids, and update budgets are frozen in the signed analysis manifest before the first definitive fit. The unit of replication and all missing or failed runs are reported. A failed run is not silently replaced using a test-informed seed.

Computational cost is separated into validation search and locked-configuration fitting. Resource use is summarized by wall-clock training time, accelerator-hours, and peak accelerator memory. Model size is reported as total and trainable parameter counts, while data exposure is reported as effective passes through unique training examples. Costs include every method-specific stage, including both cRT stages, but exclude the one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

2.7 Scope and limitations

The design isolates training-distribution effects under frozen Virchow2 representations; it does not establish how end-to-end encoder fine-tuning would respond to imbalance. Constructed tails improve causal interpretability but need not reproduce referral pathways or biological prevalence. A shared T controls data volume at the cost of discarding some available training examples, which is why the full natural condition is retained as a descriptive anchor. Some methods change more than one mechanism, so the study supports method-level comparisons rather than universal causal claims about sampling or loss design. The locked update budget equalizes optimizer steps but not information exposure: RankMix consumes teacher and student passes, MDE draws a natural and a balanced minibatch per update, and post-hoc logit adjustment performs no gradient optimization at all, so method comparisons are made at equal update budget with exposure and cost reported alongside, not at equal data throughput. MIL instance counts likewise vary with the eligible tissue area of each selected slide; processed instances, memory, and runtime are therefore reported rather than equalized by truncating bags. The number of independent split units, not the number of patches, limits inference. For BRACS in the WSI regime this limit is severe enough to be load-bearing rather than incidental: independent slide support for the scarcest classes caps the achievable severity ratio near 1.35 regardless of the shared training total, and one of its three split-unit partitions collapses the requested moderate and severe conditions to the balanced allocation entirely. Under the split-validity requirement above, BRACS WSI is therefore excluded from the confirmatory imbalance-deficit and recovery analysis and retained only as a descriptive record of this support ceiling. BRACS patch classification remains descriptive for a related reason: its patch-micro effective support, the Kish effective count of Equation (12), stays below the required threshold even under the split-unit bootstrap’s independently drawn patient weights, so its patch-level findings are reported but not entered into the confirmatory family. All imbalance-robustness findings are additionally conditional on the geometry of a single frozen encoder; whether the same deficits and recoveries hold under a different foundation representation is outside the present scope. Finally, calibration on a natural held-out cohort is conditional on that cohort’s prevalence and should not be transported to a new deployment population without explicit prior-shift analysis.

3 Evaluation and Statistical Analysis

Evaluation proceeds from description to explanation. It first measures which classes are predicted correctly, then reports PANDA-specific ordinal error and examines probability quality. It next determines whether imbalance caused a deficit, quantifies split-unit-clustered uncertainty, and tests recovery while accounting for multiple comparisons. Finally, RQ3 explores why damage and recovery differ across controlled comparisons. All endpoints, decision rules, and inferential families are fixed before test evaluation.

3.1 Discrimination endpoints and aggregation units

The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Overall accuracy is descriptive. Patch predictions are summarized in two ways: patch-micro estimates treat eligible patches as predictions while uncertainty remains split-unit-clustered, and slide/case-macro estimates first aggregate the metric contribution within each slide or split unit so that heavily tiled slides do not dominate. WSI outcomes are slide-level with patient clustering where patients contribute multiple slides and WSI/case clustering otherwise.

3.2 Ordinal outcomes for PANDA

Balanced accuracy treats every wrong class as an error of the same kind. PANDA WSI prediction is the only ordinal task in the benchmark, so it also receives two dataset-specific secondary outcomes that use the distance between ISUP grades. These measures complement rather than replace the common discrimination endpoints.

Let N be the number of held-out biopsies and C the number of grades. Let O_{ab} be the observed confusion-matrix count for true grade a and predicted grade b . The count expected under independent true and predicted grades is $E_{ab} = O_{a.}O_{.b}/N$, which retains the observed marginal grade frequencies. With $a, b \in \{0, \dots, C-1\}$ and quadratic disagreement weight $w_{ab} = (a-b)^2/(C-1)^2$, quadratic-weighted kappa is

$$\kappa_{\text{QW}} = 1 - \frac{\sum_{a=0}^{C-1} \sum_{b=0}^{C-1} w_{ab} O_{ab}}{\sum_{a=0}^{C-1} \sum_{b=0}^{C-1} w_{ab} E_{ab}}. \quad (5)$$

A value of 1 denotes perfect grade agreement, 0 is the agreement expected from the marginal grade frequencies, and negative values indicate more weighted disagreement than expected by chance. Ordinal mean absolute error (MAE) is

$$\text{MAE}_{\text{ord}} = \frac{1}{N} \sum_{i=1}^N |g_i - \hat{g}_i|, \quad (6)$$

where g_i and $\hat{g}_i$ are the true and predicted ISUP grades. It reports the average distance in grade levels and is lower-is-better. Kappa emphasizes agreement beyond the observed marginals, whereas MAE has the direct interpretation of grades missed per biopsy.

3.3 Probability quality and calibration

Discrimination metrics depend mainly on which class receives the highest score. Probability-quality metrics instead evaluate the full predicted distribution and whether confidence is compatible with observed outcomes. Tail-group macro negative log likelihood (NLL) is the co-primary calibration endpoint used by the recovery gate in Section 3.4. Natural-prevalence NLL, macro NLL, the Brier score, expected calibration error (ECE), and reliability diagrams provide complementary secondary views.

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \dots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0, 1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i .

Table 4 fixes which score variant enters each endpoint. For ordinary CE, post-hoc logit adjustment, and train-time logit adjustment, the target-prior variant is the correction defined in the companion methods report, using the natural-validation prior for the corresponding test split. For every other method, the target-prior variant equals its raw logits.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{i, y_i}. \quad (7)$$

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i, y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\text{NLL}_c = -\frac{1}{N_c} \sum_{i: y_i=c} \log \hat{p}_{ic}, \quad \text{NLL}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{NLL}_c, \quad (8)$$

Endpoint or output	Score variant
Balanced accuracy and all other discrimination endpoints	Balanced-decision logits: $z^{\text{CE}} - \tau \log \hat{\pi}$ for post-hoc logit adjustment; raw model logits for train-time logit adjustment and every other method.
Tail-NLL deficit gate and tail-NLL recovery	Target-prior-corrected probabilities before temperature scaling.
Reported NLL, Brier score, ECE, and reliability diagrams	Target-prior-corrected probabilities before temperature scaling; raw probabilities are retained as a separately labelled diagnostic.
Temperature-scaled calibration outputs	Target-prior-corrected logits divided by one positive temperature fitted on natural-validation NLL and applied unchanged to test; these secondary outputs do not enter either recovery gate.

Table 4: Canonical mapping from evaluation endpoints to score variants.

where N_c is the number of held-out predictions from class c . Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{p}}_i - Y_i\|_2^2. \quad (9)$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg \max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\text{ECE} = \sum_{b=1}^B \frac{|I_b|}{N} \left| \frac{1}{|I_b|} \sum_{i \in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|} \sum_{i \in I_b} q_i \right|, \quad (10)$$

where I_b contains the predictions assigned to bin b . ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and split-unit weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. The output semantics and prior-correction formulas of individual methods are defined in the companion methods report (Queisler, 2026b).

3.4 From imbalance deficit to mitigation recovery

Recovery is meaningful only when imbalance has first caused a measurable loss. The analysis therefore proceeds in two stages: compare size-matched balanced and imbalanced CE, then evaluate mitigation only on an endpoint that shows sufficiently large and statistically supported harm. One *comparison unit* is a fixed dataset, prediction regime, imbalance severity, and tail assignment. A *gate* is the prespecified eligibility rule that routes such a unit to the relevant recovery analysis; it is not a learned model component.

For a higher-is-better metric M , define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$D_M = M(\text{balanced CE}) - M(\text{imbalanced CE}),$$

$$R_M = \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \quad (11)$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T , methods that only reweight or recalibrate the imbalanced manifest cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M .

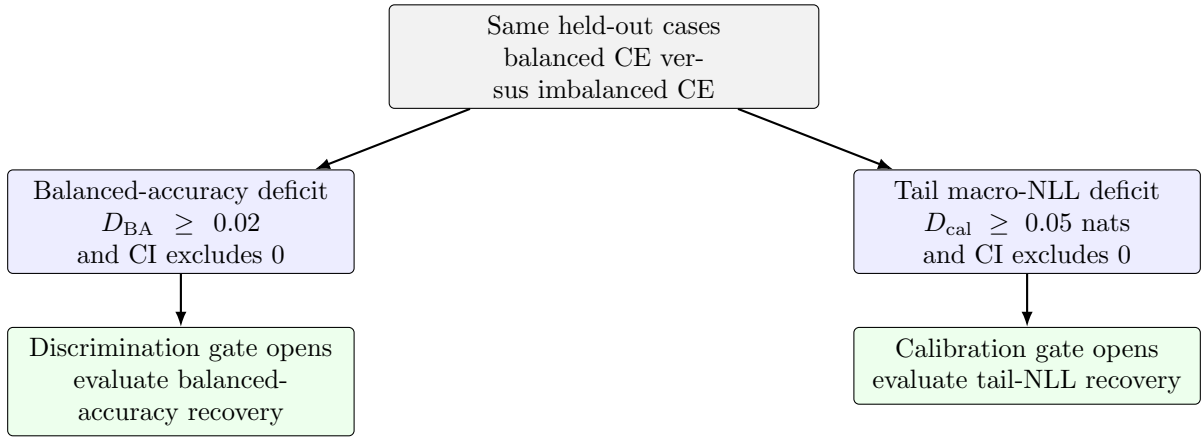


Figure 4: Routing from an imbalance deficit to mitigation analysis. A comparison unit can enter through either gate or both. If neither rule is met, its CE deficits are still reported, but a recovery ratio is not interpreted. CI denotes the paired 95% confidence interval.

The *discrimination gate* opens when the CE balanced-accuracy deficit is at least 0.02 and its paired 95% confidence interval excludes zero. The *calibration gate* opens when the CE tail-group macro-NLL deficit, $NLL_{tail}(\text{imbalanced CE}) - NLL_{tail}(\text{balanced CE})$, is at least 0.05 nats with a paired 95% confidence interval excluding zero. Both rules are computed from CE runs on the same held-out cases before any mitigation method is compared. A comparison unit entering through the discrimination gate is evaluated for balanced-accuracy recovery; one entering through the calibration gate is evaluated for tail-calibration recovery; it may enter through both.

The two routes preserve an important distinction. Strong frozen features may leave balanced accuracy nearly intact while imbalance still biases tail probabilities, in which case a single discrimination rule would exclude methods designed to correct class priors. Conversely, the gates avoid unstable recovery ratios when no meaningful damage exists to recover. All CE deficits on both axes remain reported, including units that fail both rules. For the lower-is-better calibration axis, the sign-corrected deficit $D_{cal} = NLL_{tail}(\text{imbalanced CE}) - NLL_{tail}(\text{balanced CE})$ is the denominator and the method’s tail-NLL improvement over imbalanced CE is the numerator, so Equation (11) retains the same interpretation. Recovery for other secondary metrics is descriptive and does not reopen a comparison unit that fails both gates.

3.5 Uncertainty from split-unit resampling

A *bootstrap* approximates sampling uncertainty by repeatedly constructing hypothetical test cohorts from the observed split units: patients or participants where available, and the biopsy/WSI for PANDA and CAMELYON16. Each hypothetical cohort reweights the observed split units rather than deterministically repeating each one exactly once; the resampling mechanism for split units and for confirmation seeds is detailed below. The metric is recomputed for every replicate, and the spread of those values shows how much the result could change under a similar case sample. Split units—not patches—are resampled because patches from one case are correlated.

Label-only preflight. Before model training, 10,000 bootstrap replicates are constructed using only split-unit identifiers, split membership, and class labels. *Label-only* means that no predictions, performance values, or method identities are used; these replicates test whether the planned resampling scheme is valid before results can influence it. The count of 10,000 is a prespecified Monte Carlo budget. It places about 250 replicates in each 2.5% tail of a percentile interval, reducing simulation noise in the interval endpoints while remaining inexpensive at this label-only stage.

Each split unit draws an independent, continuous bootstrap weight in every replicate: a Bayesian bootstrap (Rubin, 1981) in which the n observed units’ weights are n times a draw from a symmetric Dirichlet($1, \dots, 1$) distribution over the n -simplex. This draw is almost surely strictly positive in every coordinate, so every unit contributes to every replicate; no split–class cell, including one contributed by a unit whose observed split-by-class pattern is unique among the cohort, can be emptied by an unlucky draw. A unit’s observed contribution pattern—how many examples it supplies to each class in each test-split appearance—is still recorded and grouped into strata, but only to report per-stratum diagnostics; it no longer constrains the weight draw itself. A selected unit’s weight is applied to all of its slides and patches in every split. Figure 5 summarizes the workflow.

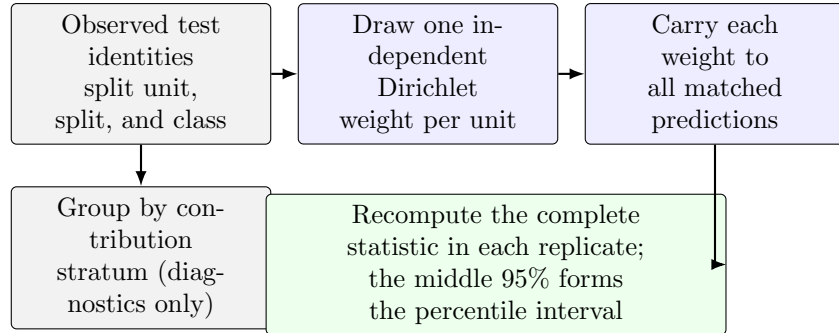


Figure 5: Split-unit-clustered Bayesian bootstrap workflow. The label-only preflight executes the identity, weight-draw, and stratum-diagnostic steps before predictions exist. After model fitting, the same paired per-unit weights are applied to every matched run.

The preflight verifies that every original split–class cell remains represented, every split-level metric is computable, one split unit receives the same weight across all of its split appearances, and every unit’s resampled weight actually varies across replicates. Failure of any check stops the analysis rather than causing a replicate to be silently discarded.

The independent per-unit weight draw can still produce a replicate dominated by a few heavily weighted units. If split unit i has class-specific bootstrap weight v_i , the *Kish effective split-unit count*

$$n_{\text{Kish}} = \frac{(\sum_i v_i)^2}{\sum_i v_i^2}. \quad (12)$$

translates unequal case weights into the equivalent number of equally weighted split units. It decreases as weight concentrates. The preflight also records each split/class cell’s number of contributing units and its largest unit-weight share. If $n_{\text{Kish}} < 5$ in any replicate, one unit supplies more than 50% of a class’s weight in more than 5% of replicates, or a split/class cell has fewer than five contributing units, inference for that dataset–regime is designated descriptive before training. The diagnostic report and resampling seed are frozen with the analysis manifest.

Intervals after model fitting. The three split-unit partitions are fixed design repetitions and are averaged with equal weight; they are not treated as three independent cohorts. Tail assignments remain separate unless an assignment-averaged estimand is explicitly reported. Within each case-bootstrap replicate, the five matched confirmation-seed indices are also resampled as paired blocks and averaged before the splits are combined. This captures algorithmic variation without pretending that five model initializations create five additional case cohorts.

Balanced CE, imbalanced CE, and each mitigation method receive the same split-unit multiplicities and confirmation-seed draws. The complete deficit and recovery fraction, including numerator and denominator, is recomputed in each replicate. The reported 95% percentile interval spans the 2.5th to 97.5th percentiles of the resulting distribution. Patches are never bootstrapped independently.

3.6 Paired testing and multiplicity control

Confidence intervals and raw effect sizes are the main evidence. Hypothesis tests provide a supporting check for the prespecified primary methods. On the discrimination axis, the tested effect is the balanced-accuracy improvement over imbalanced CE. On the calibration axis, it is the reduction in tail-group macro NLL, oriented so that positive values indicate improvement. Recovery fractions are normalized effect summaries, not test statistics.

The raw two-sided p -value comes from a paired split-unit-block permutation test. This test represents the null hypothesis of no method difference by repeatedly swapping the method and imbalanced-CE labels within a split unit. The same swap is used for all of that unit’s slides, patches, split appearances, and matched confirmation seeds, preserving the paired design. All possible swaps are enumerated when feasible. Otherwise, 100,000 random swaps are used with a plus-one correction; this gives a minimum attainable Monte Carlo p -value of approximately 10^{-5} and avoids reporting a simulated value of zero.

A *test family* is the set of related hypotheses interpreted together. Testing many hypotheses increases the chance that at least one small p -value occurs by chance. Holm adjustment controls this family-wise error by ordering the raw p -values from smallest to largest, applying the strongest correction to the smallest value, and then proceeding stepwise with less stringent corrections. It provides this control without assuming that the tests are independent.

The confirmatory family contains weighted CE, focal loss, train-time logit adjustment, and balanced sampling. These four primary methods directly target tail support or class priors and are shared across both prediction regimes. Within one dataset and regime, the family includes every gate-passing combination of primary method, gate, severity, and tail assignment; severities are not split into separate families. Holm adjustment is applied across the testable comparisons, while gated-out comparisons are recorded as not tested. All remaining methods and all secondary classwise, calibration, and cost analyses are exploratory. They retain the same effect estimates and confidence intervals but remain outside the confirmatory correction.

3.7 Exploratory explanation of damage and recovery (RQ3)

RQ1 and RQ2 estimate matched effects within a dataset and task. RQ3 takes one step back and asks why some of those controlled comparisons show more damage or recovery than others. It does not predict individual patient outcomes and it does not rank mitigation methods. For

damage occurrence and magnitude, one observation is a dataset, prediction regime, imbalance severity, and tail assignment. Recovery observations additionally identify the mitigation method and the discrimination or calibration gate through which the comparison entered.

Outcomes to explain. The analysis is divided into three linked questions rather than one opaque combined model:

1. **Damage occurrence:** did the comparison pass either prespecified deficit gate? Every controlled comparison contributes to this binary analysis.
2. **Damage magnitude:** how large was the signed balanced-accuracy deficit? Every comparison contributes, including those with no damage or an apparent improvement under imbalance.
3. **Mitigation headroom:** how much of the demonstrated deficit was recovered? Only gated comparisons contribute because recovery has no stable interpretation when there is no deficit to recover.

These three outcomes correspond directly to the sequence introduced in RQ3: occurrence, magnitude, and conditional recovery.

Candidate explanations. Only quantities available without using test outcomes are eligible as predictors. Table 5 separates the two primary predictors from the variables used only in sensitivity analyses.

Predictor	Question it represents	Model role
Achieved imbalance ratio	How extreme was the realized head–tail allocation?	Primary, log-transformed
Support–difficulty alignment $A = \text{corr}(\log N_c, d_c)$	Does greater allocated support go to more difficult classes ($A > 0$), or does scarcity compound difficulty ($A < 0$)?	Primary
Intrinsic separability	How distinguishable are the classes in a common balanced frozen-feature reference, before condition-specific support is considered?	One-predictor sensitivity
Condition-specific learnability	What remains learnable from the exact controlled training manifest?	One-predictor sensitivity
Minimum independent support	How many distinct split units or slides support the scarcest class?	One-predictor sensitivity
Effective patch support	How much patch support remains after accounting for within-case redundancy?	Patch-only sensitivity
Prediction regime	Does the comparison use patch classification or WSI-level MIL?	One-predictor sensitivity

Table 5: RQ3 predictors and their prespecified roles. Sensitivity predictors are fitted one at a time rather than added to the primary model.

Measuring separability and learnability. Intrinsic separability holds the evidence source constant. It uses one fixed balanced reference subset per dataset, regime, and split to ask whether Virchow2 already places the classes in distinguishable regions. Condition-specific learnability changes the reference to the exact balanced or imbalanced training manifest and asks how much class information remains accessible under that condition’s realized support. The distinction is between inherent feature geometry and the evidence actually retained by one constructed condition.

Both quantities are evaluated on the unchanged natural-validation cohort. Balanced k -nearest-neighbour (kNN) macro recall labels each validation embedding from its closest reference embeddings and gives every class equal weight. A *linear probe* is a class-balanced linear

classifier fitted on the frozen reference embeddings; its macro recall is the primary separability or learnability summary. Per-class nearest-neighbour error is reported descriptively to show which classes overlap. None of these diagnostics uses test labels or CE test performance. Patch probes operate on patch embeddings. WSI probes use the mean and componentwise standard deviation of a slide’s eligible instances, avoiding a learned attention summary from the model being explained.

Sensitivity to correlated patches. Raw patch count can overstate evidence when many patches come from the same case. Unique split-unit and slide counts therefore remain the primary support measures. As a secondary sensitivity analysis, within-case redundancy is summarized by the intraclass correlation (ICC) of a cross-fitted class margin. An ICC near zero means that patches from the same case are no more alike than patches from different cases; an ICC near one indicates strong redundancy.

For class c , let G_c be the number of split-unit or slide clusters, m_{gc} their sizes, and $N_c = \sum_g m_{gc}$. With between-cluster and within-cluster mean squares $MS_{B,c}$ and $MS_{W,c}$, the unequal-cluster-size correction and ICC are

$$m_{0,c} = \frac{N_c - \sum_{g=1}^{G_c} m_{gc}^2 / N_c}{G_c - 1}, \quad ICC_c = \frac{MS_{B,c} - MS_{W,c}}{MS_{B,c} + (m_{0,c} - 1)MS_{W,c}}. \quad (13)$$

The estimate is clipped to $[0, 1]$. The corresponding effective sample size is

$$N_{\text{eff},c} = \frac{N_c}{1 + (\bar{m}_c - 1) ICC_c}, \quad (14)$$

where $\bar{m}_c$ is the arithmetic mean cluster size. The scalar margin is the fixed intrinsic linear probe’s logit for class c minus its largest competing logit. It is evaluated by *cross-fitting*: each observation is scored only by a probe trained without it. Equations (13)–(14) provide a patch-only sensitivity measure, not a replacement for independent-unit counts. Slide count is used directly for WSI targets, with patient count also examined when patients contribute multiple slides.

Three deliberately small association models. One model is fitted for each of the three RQ3 outcomes. The occurrence model handles the yes/no gate result; the magnitude and recovery models handle continuous outcomes. Every primary model contains exactly two predictors: the log achieved imbalance ratio and support–difficulty alignment A . The magnitude model also uses each deficit’s bootstrap standard error so that a noisy estimated deficit is not treated as an exact observation.

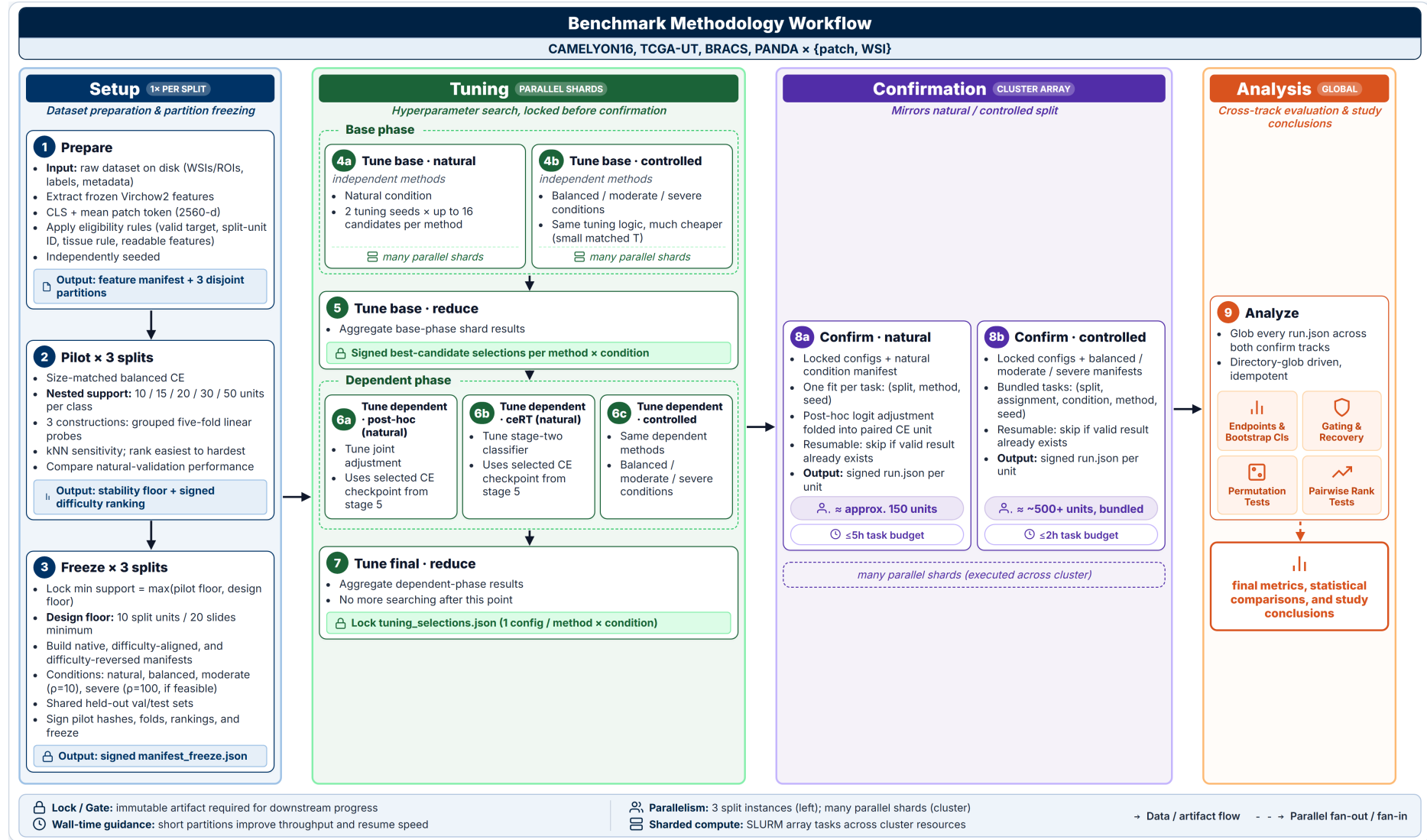
Comparisons from the same dataset–target group can share a baseline level of difficulty. A *random intercept* allows each group to have its own starting level while estimating common associations with imbalance and alignment. This adjustment is important because the analysis pools several severities and assignments from the same group. The recovery model omits method-specific effects because the small number of gated units cannot support them.

The remaining predictors in Table 5 are fitted one at a time in separate sensitivity models. They are not combined with one another or with the two primary predictors because only a few independent dataset–target groups are available and the candidate predictors are strongly confounded. No interactions are fitted.

Parameters are estimated by maximum a posteriori (MAP) fitting: ordinary model fit is combined with weak zero-centered Gaussian priors that discourage extreme slopes and variance estimates in this small sample. A leave-one-dataset–target-group-out check then refits the magnitude model while withholding one complete group at a time. This is only a coarse stability check, not evidence of predictive generalization. TCGA-UT supplies the only genuine paired

comparison between prediction regimes; the CAMELYON16, PANDA, and BRACS targets differ by regime. Model definitions, predictor roles, and priors are frozen before recovery outcomes are inspected, and all RQ3 conclusions remain hypothesis-generating.

A Benchmark Overview



B Experimental Controls

Every trainable method draws its learning rate from the common envelope

$$\eta \in \{1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}, 1 \times 10^{-2}\}, \quad (15)$$

searched through a four-value active window that starts at the envelope’s current-centered values $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}$. Focal loss, CE + soft F1, and CE + soft MCC additionally draw their auxiliary strength from the seven-value envelopes listed in Table 6, searched the same way through a four-value active window; a strength of zero is excluded from the searched window and instead serves as the CE-anchor point described in Section 2.6. Every other method’s imbalance-specific control in Table 6 uses its listed four-value grid without an envelope. For a method with an active strength window, that window is crossed with the active learning-rate window; for a method without one, the learning-rate window is searched alone or crossed with the fixed grid in Table 6. Either way the result is at most 16 candidate configurations per method, dataset, regime, and severity in any one search round. When a window’s winning value is an edge value, the window shifts by one envelope position, and only the newly added candidates are fitted; the three overlapping candidates already fitted in an earlier round are always reused, never retrained. This can repeat for at most three rounds on the learning-rate axis, three rounds on focal loss’s strength axis, and two rounds on the soft-F1/soft-MCC strength axis, bounded by each envelope’s length. Every candidate fitted, every window shift, and the resolved-or-tuning-limited outcome of each axis are completed and recorded in the signed analysis manifest before the corresponding method enters confirmation. No test data and no confirmation-stage result ever informs a shift, and no grid is changed once confirmation training begins.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0, 0.5, 1, 1.5, 2, 4, 8\}$ envelope; active window starts at $\{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0, 0.25, 1, 4, 16, 64\}$ envelope; active window starts at $\{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
RankMix-inspired	beta shape $\alpha \in \{0.5, 1, 2, 4\}$
SC-MIL	temperature $\tau \in \{0.05, 0.1, 0.5, 1\}$
MDE-inspired	consistency weight $\lambda_{\text{con}} \in \{0, 0.1, 0.25, 0.5\}$

Table 6: Prespecified method-specific grids crossed with the active window of the learning-rate envelope in Equation (15). Focal loss and the two soft-label hybrids search their listed envelope through an adaptive four-value window, shifting as described in Section 2.6; every other row is a fixed four-value grid. cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately.

For every realized condition, the experiment record contains dataset version; eligibility rules; split-unit, slide, and patch identifiers by partition; requested and achieved T and ρ ; class assignment; all seed families and their tuning, confirmation, pilot, or definitive roles; pilot patch quota; bootstrap-preflight diagnostics; model and optimizer configuration; candidate grid; selected checkpoint; environment identifier; and test-prediction hash. Deviations are dated and justified before the affected test labels are accessed.

C Terminology

Natural anchor. The full eligible natural training partition. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M .

Imbalance severity. The requested training allocation summarized by ρ ; achieved severity is computed from realized integer counts.

Pilot level. A candidate number of independent training units per class used in the balanced CE support-stability study.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Comparison unit. One dataset, prediction regime, imbalance severity, and tail assignment evaluated together.

Deficit gate. A prespecified threshold rule that determines whether an observed discrimination or calibration deficit is large and precise enough for recovery analysis.

Imbalance deficit. The paired balanced-CE minus imbalanced-CE difference at common T and on the same held-out cases.

Recovery. A method’s improvement over imbalanced CE divided by the evidenced imbalance deficit.

Kish effective split-unit count. The equally weighted split-unit count with the same weight concentration as one bootstrap replicate.

Intraclass correlation. The estimated similarity of class margins from patches belonging to the same split-unit or slide cluster.

Patch-micro estimate. A metric computed over eligible patch predictions, with split-unit-clustered uncertainty.

Case-macro estimate. An equal-weight aggregation over slides or split units, limiting domination by heavily tiled cases.

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